using Microsoft's Visual Studio .NET environment. KMF (in PXI platform) helps to develop high-performance, scalable test measurement solutions, which take full advantage of parallel-processing multi-core CPUs for faster measurement processing and testing speed," shares Keshav Bapat, general manager at Keysight Technologies.

Arbitrary waveform generators

There is a demand for more cost-efficient waveform generators, too. Manufacturers want signal generators that can replace expensive and complex systems that are customised to test only a single device.

Test equipment manufacturers have come up with arbitrary waveform generators (AWGs) that cater to this need. AWG-based signal generation can also simplify emulation of various operational scenarios during the development of radar systems, electronic warfare systems, satellite communications systems and more.

Both Keysight Technologies and NI have developed AWGs that offer increased bandwidth, resolution and number of channels. Tektronix, on the other hand, claims that its recently developed AWG offers a substantial increase in signal consistency and strength, and a sample rate of 10GSa/s, simplifying the job of signal generation and RF engineers. The device is expected to hit the market soon.

Oscilloscopes

Testing challenges continue to grow as engineers develop systems with higher data rates. As speeds go up and amplitudes down, system noise becomes a major problem because it blocks important signal details. Therefore oscilloscopes are being refined for improved signal fidelity and channel scalability while occupying smaller footprint.

NI has already introduced in the market oscilloscopes based on PXI platform that offer up to 400MHz of analogue bandwidth and flexible measurement configurations. Tektronix is also gearing up to launch a high-frequency oscilloscope.

Other advances

These are just a few major advances made in semiconductor, signal system and transceiver based test equipment that are primarily used by defence, robotics, communication and automotive industries. According to market players, the ATE industry will continue to evolve as the requirements of test engineers grow.

Industry expert Leela Krishna, from Rohde & Schwarz, shares that sometimes the test equipment is customised to suit the particular needs of a business. "Usually, our customers from the defence sector demand high-frequency test equipment. In such a case, we tweak the software of the instrument or else try to manufacture a new hardware for the instrument."

For instance, "Traditionally, ATEs had customised pin electronic cards that were made for specific test criteria. These have now been replaced with PXI platform slots, which can be used as per the scope of work," informs Nitin Nigam, field application engineer at Tektronix.

To sum up

Design for manufacturability is the first phase of a perfect design, leading to the next phase of advancements in measuring instruments. Modular ATE is a radical transformation of the testing process in sync with the product design trends. Combining PXI platform with a modular hardware architecture allows engineers to develop scalable test systems that can meet the needs of different business segments while lowering the cost and size of test systems. So we can say it is all about a revolution leading to evolution! EFY